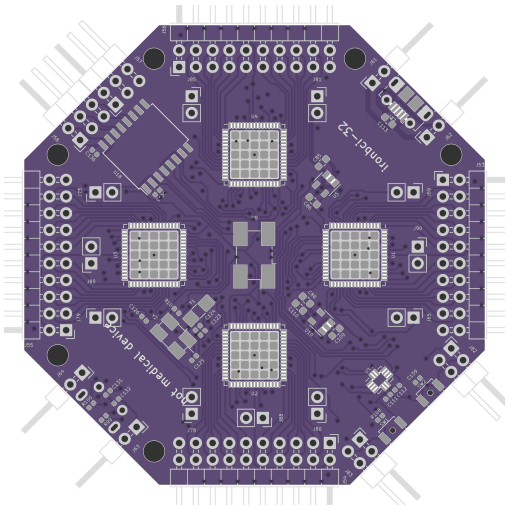
PCB TECHNICAL INFORMATION · SKU PIA24025P

Four-Layer PCB — Fabrication & Layout Data

Board-level data for manufacturing, inspection and enclosure design: stack-up, copper layers, drill table and the Gerber file set published in the IronBCI-32 repository.

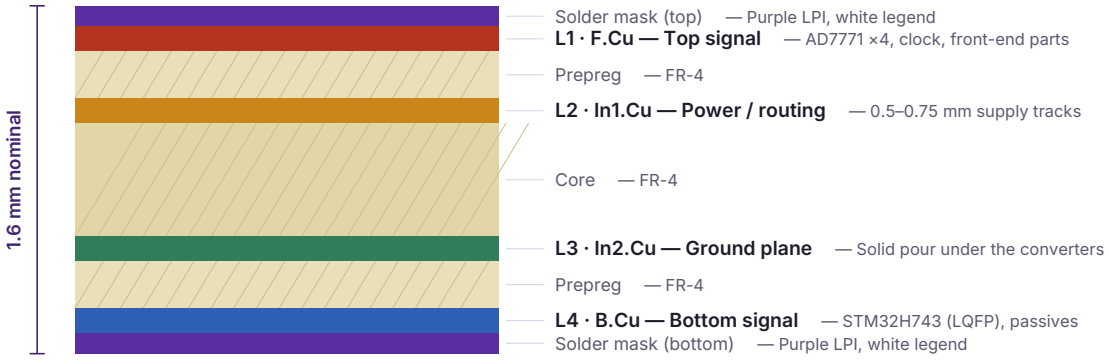


4	1.6 mm	70 × 70	0.25 mm	0.4 mm
Copper layers	FR-4 board thickness	mm octagon outline	Minimum track width	Minimum drill (via)

01 FABRICATION SUMMARY

BOARD	
Base material	FR-4
Layer count	4 copper layers (signal / power / ground / signal)
Finished thickness	1.6 mm
Outline	Regular octagon, 70.00 × 70.00 mm across flats, 8 edges of 29.00 mm
Solder mask / legend	Purple solder mask both sides, white legend (as supplied board)
COPPER FEATURES (MEASURED FROM GERBER)	
Signal tracks	0.25 mm minimum (top and bottom signal routing)
Power tracks	0.50 mm and 0.75 mm (mainly on inner layer L2)
Vias	264 × Ø0.40 mm drill, Ø0.80 mm pad (0.20 mm annular ring)
Component holes	130 × Ø1.00 mm plated (2.54 mm headers and pin connectors), 4 × Ø0.85 mm plated (USB shell tabs)
Non-plated holes	6 × Ø3.20 mm (M3 mounting), 2 × Ø0.80 mm (locating pegs)
DATA	
Design tool	KiCad 9.0.6 (Pcbnew), files generated 8 Nov 2025
Formats	Gerber X2 (RS-274X with attributes), Excellon drill, metric, absolute coordinates
Repository	github.com/pieeg-club/ironbci-32 — Gerber/ironbci_Gerber.zip, Electrical/ironbci.kicad_sch, ironbci_scheme.pdf

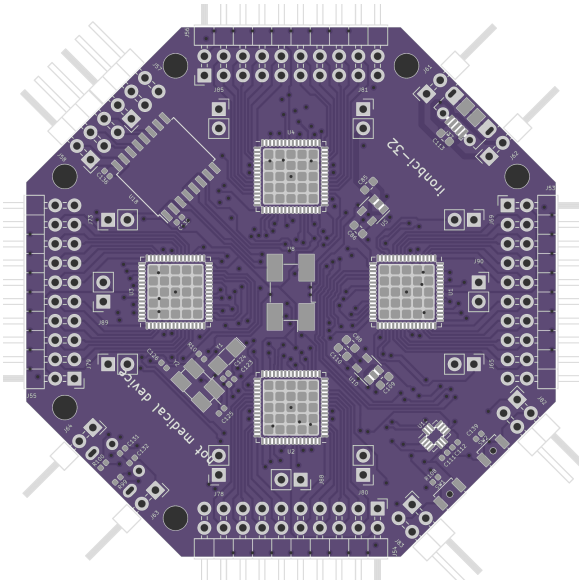
02 LAYER STACK-UP



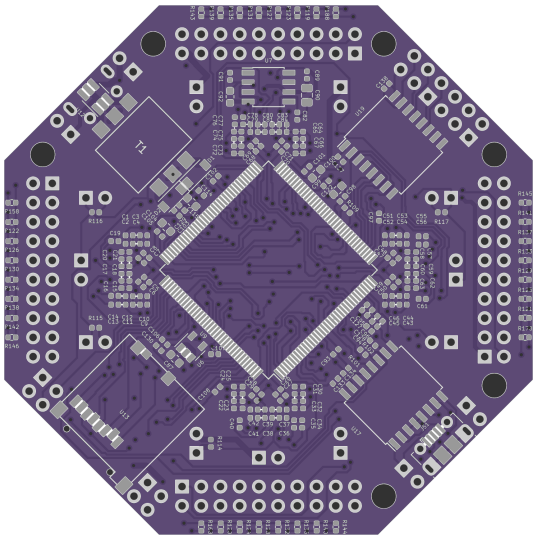
The two outer layers carry the signal routing: the four AD7771 converters, clock and analog front-end parts sit on the top side, while the STM32H743 (LQFP-144, rotated 45°) and its decoupling sit on the bottom.

Inner layer L2 distributes the supply rails with 0.5–0.75 mm tracks; L3 is a near-continuous ground pour beneath the converters, giving every analog path a short, low-impedance return. Dielectric thicknesses follow the fabricator's standard 4-layer 1.6 mm build.

03 ASSEMBLY VIEWS

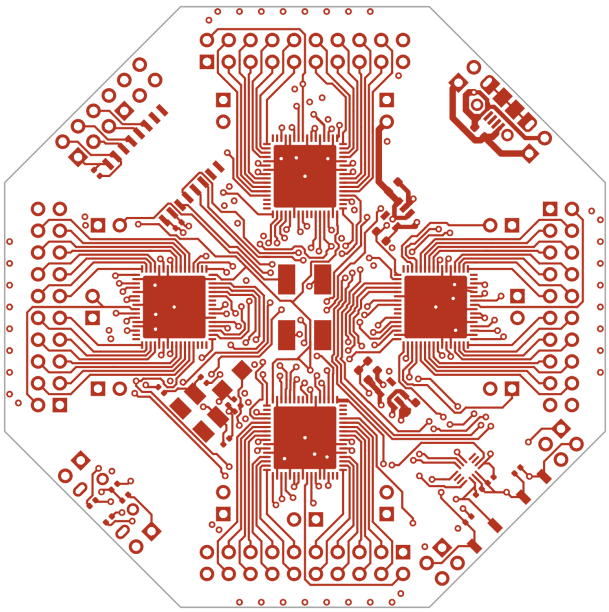


Top side — 4 × AD7771 (U1–U4) around the central clock

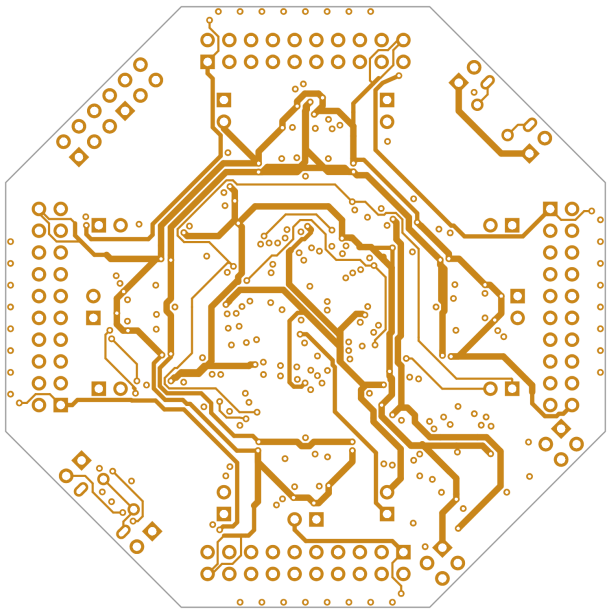


Bottom side (mirrored view) — STM32H743 LQFP-144, supply section

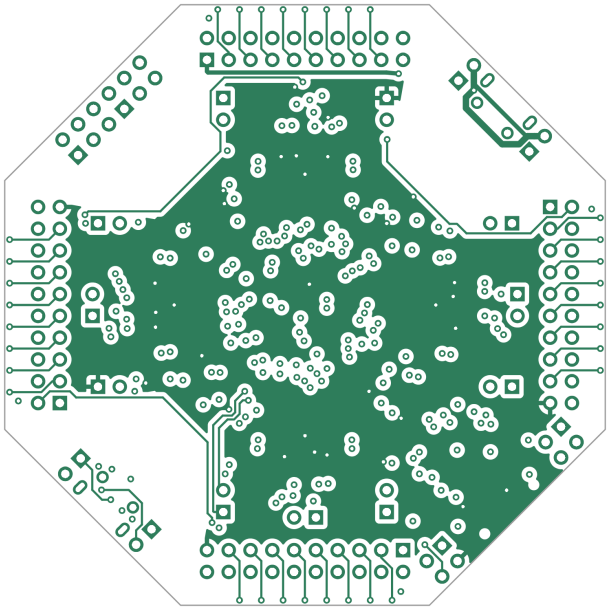
04 COPPER LAYER ARTWORK



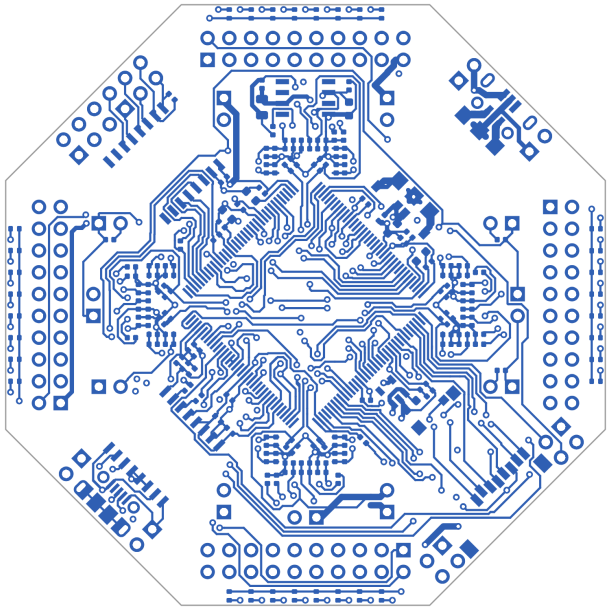
L1 · F.Cu — top signal layer



L2 · In1.Cu — power distribution



L3 · In2.Cu — ground plane



L4 · B.Cu — bottom signal layer (viewed from top)

Rendered directly from the published Gerber files at 1:1 registration; grey line = board outline, white dots = drilled holes.

For research and education only. Not a medical device. Specifications subject to change.

05 DRILL TABLE

Tool	Diameter	Plating	Qty	Function
T1	Ø0.40 mm	Plated	264	Vias (Ø0.80 mm pad)
T3	Ø0.85 mm	Plated	4	Micro-USB shell tabs
T4	Ø1.00 mm	Plated	130	2.54 mm headers, pin connectors
T1 (NPTH)	Ø0.80 mm	Non-plated	2	Locating pegs
T2 (NPTH)	Ø3.20 mm	Non-plated	6	M3 mounting holes (see IRB32-MD-002)
		Total	**406**	

06 GERBER FILE SET

File	Layer	Function
`ironbci-beta-F_Cu.gbr`	L1	Copper, top
`ironbci-beta-In1_Cu.gbr`	L2	Copper, inner 1
`ironbci-beta-In2_Cu.gbr`	L3	Copper, inner 2
`ironbci-beta-B_Cu.gbr`	L4	Copper, bottom
`ironbci-beta-F_Mask.gbr / B_Mask.gbr`	Top / Bot	Solder mask
`ironbci-beta-F_Silkscreen.gbr / B_Silkscreen.gbr`	Top / Bot	Legend
`ironbci-beta-F_Paste.gbr / B_Paste.gbr`	Top / Bot	Solder paste (stencil)
`ironbci-beta-Edge_Cuts.gbr`	—	Board profile
`ironbci-beta-PTH.drl / NPTH.drl`	—	Plated / non-plated drill (Excellon)

07 ORDERING BARE BOARDS

Step	Action
1	Download `Gerber/ironbci_Gerber.zip` from the repository and upload it unchanged to the PCB fabricator.
2	Select 4 layers, 1.6 mm FR-4, 70 × 70 mm. Check that the viewer shows the octagonal outline and all four copper layers.
3	Layer order: F_Cu → In1_Cu → In2_Cu → B_Cu. Choose purple (or any) solder mask and white legend.
4	Confirm the fabricator's capability for 0.25 mm tracks and 0.4 mm vias (standard on most 4-layer services).
5	Order a stencil for both sides if assembling in-house: the top carries the AD7771s, the bottom the STM32H743.
6	After assembly, flash firmware through SWD (see IRB32-SW-006) and check that a USB serial port appears.

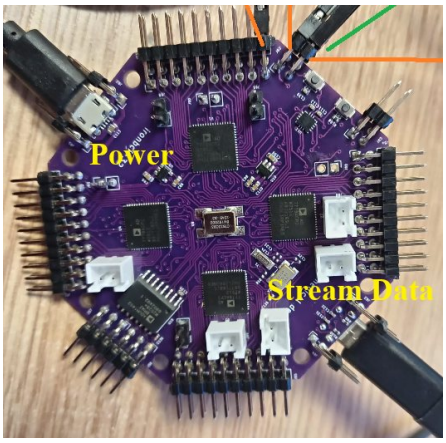
NOTE Ready-made boards.

Fully assembled and factory-tested IronBCI-32 boards are available from Elecrow, PiEEG's official manufacturing partner (SKU PIA24025P). Self-built boards are not covered by that testing.

08 ELECTRICAL DESIGN FILES

The schematic is published as a KiCad 9 project sheet (`Electrical/ironbci.kicad_sch`) and a 22-page A4 PDF (`Electrical/ironbci_scheme.pdf`). It covers the STM32H743 pin map, the four AD7771 channel groups with their header connections, clocking, supplies and the USB interfaces.

Firmware: ready-to-flash `hex/ironbci32.hex` and the STM32H743 Makefile project (GCC Arm, OpenOCD) in `Framework/` (beta).



Assembled board with Power and Data Stream cables.

09 RELATED DOCUMENTS

Doc. No.	Document	Contents
IRB32-DS-001	Product Datasheet	Features, specifications, block diagram, ordering
IRB32-MD-002	Mechanical & Dimensional Drawing	Outline, mounting holes, connector positions
IRB32-PIN-004	Connector Pinout & Electrode Guide	Headers, pin order, REF/BIAS, 10-20 layout
IRB32-QSG-005	Quick Start & Safety Guide	Setup, battery-only operation, troubleshooting
IRB32-SW-006	Data Interface & Software Guide	Serial frame, µV conversion, SDKs, firmware